Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 58190

Part A: Each question is worth 3 points

1. Basil, who is older than Peter by 3 years and 2 days, was born on October 1, 2012.
When was Peter born?

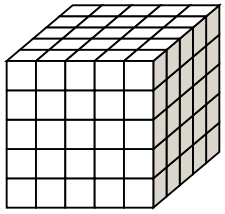
A. October 3, 2015 ***B.*** September 29, 2009 ***C.*** October 1, 2015
D. October 3, 2009 ***E.*** October 3, 2016

2. Nina builds a straight fence from 13 pieces: posts and boards take turns, and the fence starts and ends with a post.
How many posts are there?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Part B: Each question is worth 4 points

3. A cube made of 5 × 5 × 5 small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 2 painted faces?

A. 36 ***B.*** 32 ***C.*** 31 ***D.*** 30 ***E.*** 28

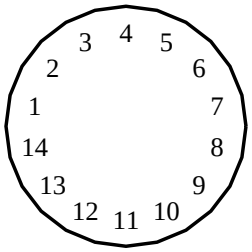
4. What is the smallest number whose digits add up to 23?

A. 698 ***B.*** 689 ***C.*** 600 ***D.*** 599 ***E.*** 487

5. Emma thinks of a number, multiplies by 3, then divides by 3, then multiplies by 2, and gets 38.
Which number did Emma think of?

A. 76 ***B.*** 38 ***C.*** 21 ***D.*** 19 ***E.*** 16

6. A round cake with the numbers 1 to 14 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



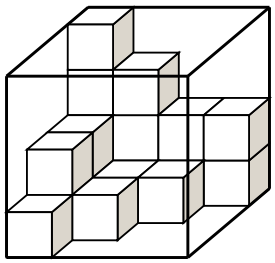
What is the largest of these sums?

A. 35 ***B.*** 43 ***C.*** 51 ***D.*** 52 ***E.*** 53

7. Only the even numbers from 2 to 120 are written down: 2, 4, 6, ..., 120.
How many digits are written?

A. 145 ***B.*** 144 ***C.*** 136 ***D.*** 129 ***E.*** 127

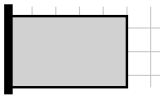
8. Mia puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Now Mia wants to cover the whole bottom of the box with one layer of cubes.



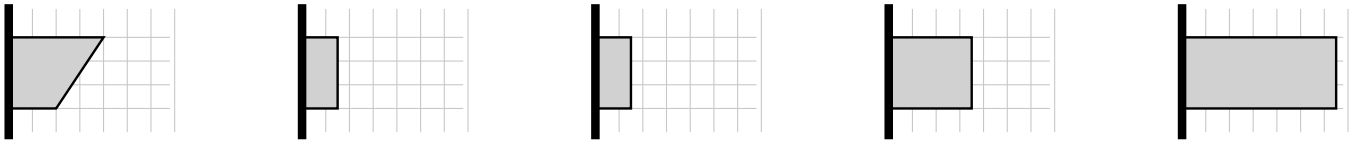
How many more cubes are needed for that?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

9. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat

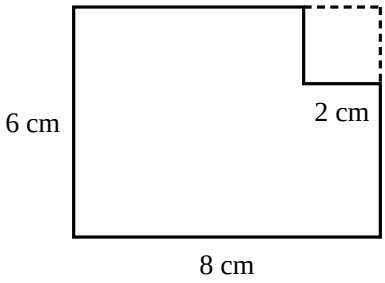


A B C D E

How many of the five drawings can be right?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. A rectangle is 8 cm long and 6 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

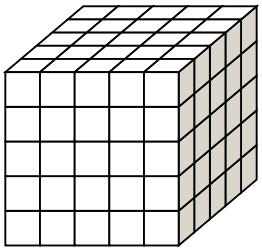
A. 36 cm ***B.*** 35 cm ***C.*** 32 cm ***D.*** 28 cm ***E.*** 20 cm

11. Tom writes the word LEMONADE. Then Tom crosses out every letter that also appears in the word GARDEN.

How many different letters are **not** crossed out?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

12. Nina glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have paint on exactly three faces?

A. 8 ***B.*** 18 ***C.*** 27 ***D.*** 36 ***E.*** 54

Part C: Each question is worth 5 points

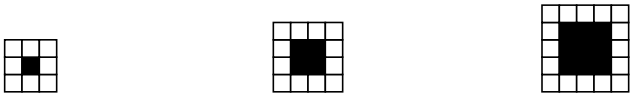
13. Aisha, Omar, Tom, Zoe and Hana each play a different instrument: the piano, the violin, the harp, the guitar and the flute. Tom uses a bow. Hana presses keys. Aisha plays an instrument with strings. Omar doesn't play the harp. Omar plays an instrument with strings.
Which statement is **not** true?

A. Zoe plays the flute ***B.*** Aisha plays the harp ***C.*** Tom plays the violin
D. Omar plays the guitar ***E.*** Omar plays the violin

14. Eva's new rug is 4 m 8 cm long. Four of these lengths are the same as that.
Which one is **not** the same?

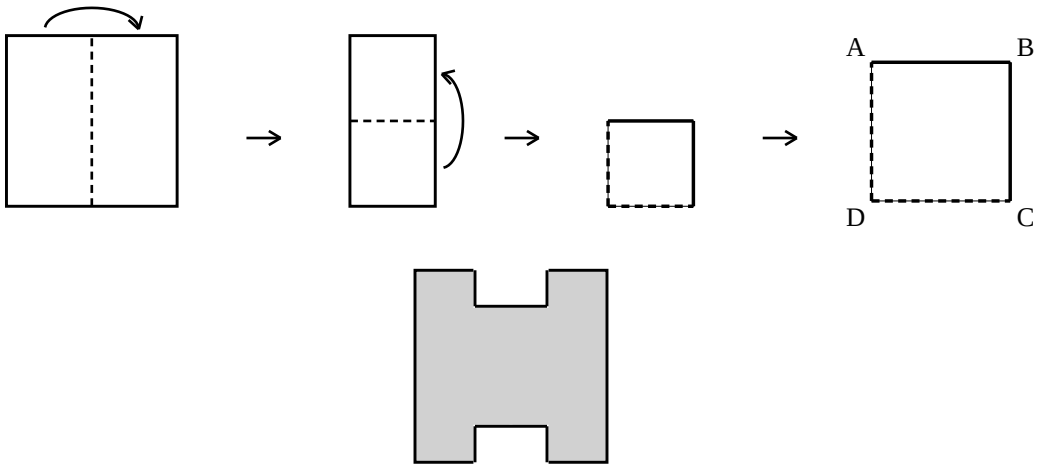
A. 4080 mm ***B.*** 408 mm ***C.*** 4 m 80 mm ***D.*** 40 dm 8 cm ***E.*** 408 cm

15. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 36 white cells. Which one is it?



A. the 5th ***B.*** the 7th ***C.*** the 8th ***D.*** the 9th ***E.*** the 12th

16. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square was cut off, and after unfolding the paper looks like the bottom picture.



Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

17. The odometer of a scooter shows 2459 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 4 ***E.*** 1

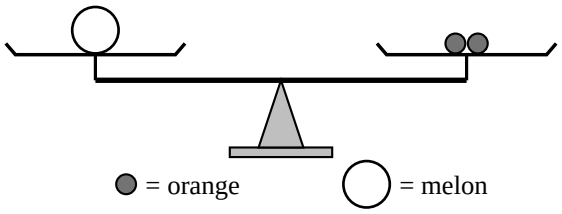
18. Martha has the cards 1, 3, 4 and 6. Using every card exactly once, with +, –, ×, ÷ and brackets, Martha wants to get the biggest possible number.
What is the biggest number Martha can get?

A. 105 ***B.*** 96 ***C.*** 84 ***D.*** 72 ***E.*** 71

19. Exactly 486 digits are used to write all the house numbers in a street, starting from 1.
How many houses are there?

A. 99 ***B.*** 162 ***C.*** 197 ***D.*** 198 ***E.*** 238

20. The balance shows that one melon weighs as much as 2 oranges. 2 melons and 2 oranges weigh 450 g altogether.



How much does one orange weigh?

A. 75 g ***B.*** 55 g ***C.*** 50 g ***D.*** 45 g ***E.*** 25 g

Answer Key

1.	**A**	**2.**	**D**	**3.**	**E**	**4.**	**D**	**5.**	**D**	**6.**	**E**
7.	**E**	**8.**	**E**	**9.**	**D**	**10.**	**D**	**11.**	**D**	**12.**	**A**
13.	**E**	**14.**	**B**	**15.**	**C**	**16.**	**A**	**17.**	**A**	**18.**	**B**
19.	**D**	**20.**	**A**								

Solutions

1. A (October 3, 2015) — Older means born earlier, so Peter was born 3 years and 2 days **after** October 1, 2012: 3 years later is October 1, 2015, then 2 more days: October 3, 2015. [2007 Q12 · younger_birthday]

2. D (7) — Post, board, post, ..., post: there is one more post than boards. 13 = 7 + 6, so 7 posts and 6 boards. [2003 Q17 · fence]

3. E (28) — Top edges without their corners: 4 × 3 = 12 (top and one side). Upright edges without the top corner: 4 × 4 = 16 (two sides; the bottom corners count here because their bottom is not painted). Total 28. [2007 Q17 · not_underneath]

4. D (599) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 599: 5 + 9 + 9 = 23. [2007 Q4 · smallest_with_sum]

5. D (19) — Go backwards: 38 ÷ 2 = 19, 19 × 3 = 57, 57 ÷ 3 = 19. [2007 Q5 · think_of_a_number]

6. E (53) — All the numbers add up to 105. Two consecutive numbers that add up to 105: 52 + 53 = 105. The largest is 53. [2002 Q11 · largest_sum]

7. E (127) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 11. 4 × 1 + 45 × 2 + 11 × 3 = 127. [2007 Q23 · even_only]

8. E (6) — The bottom has 4 × 4 = 16 places. 10 of them already have a column of cubes. 16 – 10 = 6. [2007 Q14 · bottom_layer]

9. D (4) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 5 squares from the pole. Wrong: E. So 4 can be right. [2002 Q15 · count_possible]

10. D (28 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (8 + 6) = 28 cm. [2002 Q6 · corner_cut]

11. D (3) — Crossed out: A, D, E and N (they are in GARDEN). Left: L, M and O, so 3 different letters. [2007 Q3 · only_in_first]

12. A (8) — They are the corners, and a cube always has 8 corners. [2003 Q9 · painted]

13. E (Omar plays the violin) — The only possible matching: Aisha – harp; Omar – guitar; Tom – violin; Zoe – flute; Hana – piano. [2002 Q18 · which_not_true]

14. B (408 mm) — 4 m 8 cm = 408 cm = 4080 mm. But 408 mm = 408 mm. [2003 Q10 · not_equal]

15. C (the 8th) — Try the diagrams one by one: the 8th is 10 × 10 = 100 cells with 64 black, so 36 white. [2007 Q21 · which_diagram]

16. A (corner A) — The design shows notches in the middle of the top and bottom sides. That comes from cutting corner A (where one fold meets a paper edge). [2007 Q24 · which_corner]

17. A (8) — Keep as many digits on the left as possible. The next such number is 2467: 2467 – 2459 = 8. [2002 Q24 · going_up]

18. B (96) — (1 + 3) × 4 × 6 = 96. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

19. D (198) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 486 – 189 = 297 digits are three-digit numbers: 297 ÷ 3 = 99, from 100 to 198. [2003 Q16 · three_digit_pages]

20. A (75 g) — Swap each melon for 2 oranges: 2 × 2 + 2 = 6 oranges weigh 450 g. One orange: 450 ÷ 6 = 75 g. [2002 Q9 · grams]